

Lumen — Portfolio Documentation

AI-Based Web Accessibility Checker

Repository: github.com/balisikh/ai-based-web-accessibility-checker · Local:
<http://localhost:4376>

Assistive findings only — not a legal accessibility certificate or formal conformance audit.

Overview

Lumen scans a public URL with Playwright + axe-core, returns a score, severities, rule help, optional AI tips, and JSON export.

1. Paste a public website URL
2. Headless Chrome renders the page
3. axe-core checks WCAG A/AA-oriented rules
4. Score, severities, and issue details
5. Optional AI tips; JSON export

UI screenshots

Batch results — 28 sites

Week 10: Week 10 results		Week 11: Week 11 results		Week 12: Week 12 results		Week 13: Week 13 results		Week 14: Week 14 results		Week 15: Week 15 results		Week 16: Week 16 results		Week 17: Week 17 results		Week 18: Week 18 results		Week 19: Week 19 results		Week 20: Week 20 results		Week 21: Week 21 results		Week 22: Week 22 results		Week 23: Week 23 results		Week 24: Week 24 results		Week 25: Week 25 results		Week 26: Week 26 results		Week 27: Week 27 results		Week 28: Week 28 results		Week 29: Week 29 results		Week 30: Week 30 results		Week 31: Week 31 results		Week 32: Week 32 results		Week 33: Week 33 results		Week 34: Week 34 results		Week 35: Week 35 results		Week 36: Week 36 results		Week 37: Week 37 results		Week 38: Week 38 results		Week 39: Week 39 results		Week 40: Week 40 results		Week 41: Week 41 results		Week 42: Week 42 results		Week 43: Week 43 results		Week 44: Week 44 results		Week 45: Week 45 results		Week 46: Week 46 results		Week 47: Week 47 results		Week 48: Week 48 results		Week 49: Week 49 results		Week 50: Week 50 results		Week 51: Week 51 results		Week 52: Week 52 results		Week 53: Week 53 results		Week 54: Week 54 results		Week 55: Week 55 results		Week 56: Week 56 results		Week 57: Week 57 results		Week 58: Week 58 results		Week 59: Week 59 results		Week 60: Week 60 results		Week 61: Week 61 results		Week 62: Week 62 results		Week 63: Week 63 results		Week 64: Week 64 results		Week 65: Week 65 results		Week 66: Week 66 results		Week 67: Week 67 results		Week 68: Week 68 results		Week 69: Week 69 results		Week 70: Week 70 results		Week 71: Week 71 results		Week 72: Week 72 results		Week 73: Week 73 results		Week 74: Week 74 results		Week 75: Week 75 results		Week 76: Week 76 results		Week 77: Week 77 results		Week 78: Week 78 results		Week 79: Week 79 results		Week 80: Week 80 results		Week 81: Week 81 results		Week 82: Week 82 results		Week 83: Week 83 results		Week 84: Week 84 results		Week 85: Week 85 results		Week 86: Week 86 results		Week 87: Week 87 results		Week 88: Week 88 results		Week 89: Week 89 results		Week 90: Week 90 results		Week 91: Week 91 results		Week 92: Week 92 results		Week 93: Week 93 results		Week 94: Week 94 results		Week 95: Week 95 results		Week 96: Week 96 results		Week 97: Week 97 results		Week 98: Week 98 results		Week 99: Week 99 results		Week 100: Week 100 results		Week 101: Week 101 results		Week 102: Week 102 results		Week 103: Week 103 results		Week 104: Week 104 results		Week 105: Week 105 results		Week 106: Week 106 results		Week 107: Week 107 results		Week 108: Week 108 results		Week 109: Week 109 results		Week 110: Week 110 results		Week 111: Week 111 results		Week 112: Week 112 results		Week 113: Week 113 results		Week 114: Week 114 results		Week 115: Week 115 results		Week 116: Week 116 results		Week 117: Week 117 results		Week 118: Week 118 results		Week 119: Week 119 results		Week 120: Week 120 results		Week 121: Week 121 results		Week 122: Week 122 results		Week 123: Week 123 results		Week 124: Week 124 results		Week 125: Week 125 results		Week 126: Week 126 results		Week 127: Week 127 results		Week 128: Week 128 results		Week 129: Week 129 results		Week 130: Week 130 results	
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Results — sample layout

L

Lumen

Checker

Batch results

Light

Dark

System

← New scan

Lumen

Results

https://www.w3.org/WAI/demos/bad/before/home.html

42

SCORE

Needs work

BATCH: FAIL

Critical: 3. 3 critical issues; score 42 (need ≥ 85)

3 critical issues — address these before lower-severity items, even if the score looks acceptable.

Score 0–100 from automated issue counts. **Needs work** means 42 (85+ is Strong). In our website test batches, **Pass** also requires zero critical issues.

CRITICAL

3

SERIOUS

2

MODERATE

5

MINOR

1

CRITICAL

Blocks many users — fix first.

SERIOUS

Major barriers for some users.

Filter severity

All

Export JSON

New scan

Assistive findings only — not a legal accessibility certificate.

CRITICAL

Images must have alternate text: decorative banner missing alt

SERIOUS

Elements must meet minimum color contrast ratio thresholds

CRITICAL

Images must have alternate text

WCAG

1.1.1

Rule

image-alt

Selector

img.hero-banner

SNIPPET

Mobile home (390px)



Layout (light mode)

- **Header:** Lumen logo, Checker / Batch results, Light / Dark / System
- **Background:** Soft grey gradient + grid pattern
- **Card:** White panel with teal gradient strip on top
- **Home CTA:** Orange Check accessibility button
- **Batch CTA:** Teal Run a live scan button

Batch snapshot (2026-07-30)

Metric	Value
Websites tested	28
Passed	8 (4 clean · 4 with issues)
Failed	20
Total issues	374

Pass rule: Score ≥ 85 and Critical = 0.

Passed sites: Google UK, BBC iPlayer, BBC News, Disney+ UK, GitHub (balisikh), Google Maps, Wikipedia, example.com

MVP features

Area	Status
Scan UI, Playwright + axe	Done
Rate limiting, persistence, JSON export	Done
Optional AI tips, anonymous use	Done
PDF export / accounts / crawl	Not yet

Stack

- Next.js (React) + TypeScript
- Playwright (Chrome) + axe-core
- Postgres or PGLite
- Optional OpenAI-compatible AI tips

Local development note — plain-text UI incident (Aug 2026)

During local testing the app briefly showed **unstyled plain text** (e.g. nav labels run together). The design was not removed — **CSS failed to load**.

Symptom: White page, black text; HTML OK but stylesheet returned **500** from a stale/corrupt `.next` build on port 4376.

Cause: On low-memory Windows, Turbopack could crash compiling `globals.css`, leaving a broken build that still served HTML.

Fix (commit 541ec96):

- `dev / build` default to **webpack**
- `web/scripts/fix-styles.ps1` — stop server, delete `.next`, rebuild, restart
- Troubleshooting in `web/README.md` — confirm CSS status **200** in DevTools → Network

Resolution: After clean rebuild, UI matched screenshots again (header, gradient, card, batch table). Images in this PDF show the intended layout.